

Matrix inequalities and norms

Nobori's spectral-middle-factor commutator conjecture

Difficulty: challenging **Importance:** interesting to the community

Provenance: explicit conjecture

Problem statement

Let $m, n \geq 2$ be integers. For every $A, C \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{n \times m}$, is

$$\|ABC - CBA\|_F^2 \leq 2\|B\|_2^2(\sigma_1(A)^2 + \sigma_2(A)^2)\|C\|_F^2?$$

Here $\|X\|_F = (\text{tr}(X^*X))^{1/2}$, $\|X\|_2 = \sigma_1(X)$, and $\sigma_1(X) \geq \sigma_2(X) \geq \dots$ are the singular values. Zero matrices and deficient ranks are allowed.

Relevance

The expression measures failure of two rectangular factors to commute through a middle map. A bound using the middle factor's operator norm and only two singular values of an outer factor would sharpen estimates for products of rectangular matrices.

References

1. M. Nobori, *A generalization of the Böttcher–Wenzel inequality for three rectangular matrices*, Linear Algebra and its Applications 725 (2025), 135–144. Conjecture 3.1, equation (8), in [arXiv:2506.17365v2](#); [journal](#).
2. M. Nobori, *A note on some upper bounds for the Frobenius norm of the q -deformed commutator*, arXiv:2608.03897v1 (2026), Theorem 1.1 and §3. [Primary manuscript](#).

Status check — 2026-09-08

The original arXiv record remains v2, dated 3 July 2025. Its proved Theorem 1.1 places the spectral norm on C and the Frobenius norm on B ; that is a different assertion. The author's August 2026 paper treats two-factor expressions $AB - qBA$, and does not claim this conjecture. Searches used *Nobori Conjecture 3.1, three rectangular matrices conjecture proof*, and *Nobori commutator spectral conjecture*. No resolution was located. The original conjecture and later theorem statements were inspected; the subscription journal body was not independently retrieved.